Diagnostic Dialogue Optimization: Repairing System Prompts through Multi-Turn Teacher–Student Interrogation

[TODO: Your Name][1]

[1]**[TODO: Your Affiliation]** , `[TODO: your.email@domain]`

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Abstract

Automatic prompt optimization (APO) has largely converged on a single template: run a candidate prompt over a labeled dataset, collect aggregate errors, and ask a language model to rewrite the prompt. This batch-and-rewrite paradigm treats the model under optimization as an opaque scoring function and discards the rich, turn-by-turn information that surfaces when one actually *converses* with it. We introduce **Diagnostic Dialogue Optimization (DDO)**, a prompt optimization framework in which a stronger *teacher* model conducts a structured, multi-turn diagnostic conversation with a *student* model that operates under the prompt being optimized. Rather than scoring the student in aggregate, the teacher interrogates it—probing reasoning, edge-case handling, format adherence, and instruction following—accumulates a persistent *weakness profile* across turns, and then issues a targeted, minimal edit to the student's system prompt that addresses the specific deficiency the conversation exposed. The conversation is then reset and the loop repeats on the repaired prompt. We formalize DDO as a partially observable optimization process, give an explicit algorithm, and distinguish it from (i) batch APO methods such as ProTeGi and OPRO, (ii) single-turn "Socratic" prompt-rewriting such as MARS, and (iii) conversational evaluation and red-teaming methods such as ATA and PAIR, which diagnose but do not repair. We describe a complete experimental protocol over reasoning, instruction-following, and tool-use benchmarks, including baselines, ablations, and metrics. **[TODO: Results to be added after experiments are run.]**

1 Introduction

The behavior of a large language model (LLM) in deployment is governed to a remarkable degree by its prompt. A small change in wording, ordering, or the inclusion of a single clarifying instruction can move task accuracy by double-digit percentages. This sensitivity has motivated a large body of work on *automatic prompt optimization* (APO): rather than relying on human trial-and-error, an algorithm searches the space of prompts to maximize task performance.

Despite a diversity of search strategies—reinforcement learning, evolutionary methods, beam search, textual "gradients"—the dominant APO paradigm shares a common structure. A candidate prompt is evaluated by running it over a held-out set of labeled examples; the resulting errors are aggregated; an optimizer LLM is shown a summary of those errors and asked to produce a revised prompt; and the cycle repeats. We call this the *batch-and-rewrite* paradigm. Representative instances include Automatic Prompt Engineer (APE) [1], Optimization by PROmpting (OPRO) [2], and Prompt Optimization with Textual Gradients (ProTeGi) [3].

The batch-and-rewrite paradigm has a structural blind spot. It treats the model under optimization as a black-box scoring function: the only signal that flows back to the optimizer is an aggregate error rate, optionally accompanied by a handful of failing input–output pairs. But anyone who has manually debugged a prompt knows that the most informative signal does not come from a score—it comes from *conversing* with the model. When a practitioner suspects a weakness, they do not run a thousand examples; they ask the model a pointed question, observe how it stumbles, ask a sharper follow-up, and within a few turns localize the exact misunderstanding. The prompt edit that follows is then surgical rather than speculative.

This paper asks: *can we automate that diagnostic conversation?* We propose **Diagnostic Dialogue Optimization (DDO)**, in which a stronger teacher model is given (i) a description of what aspects of behavior matter for the task and (ii) read access to the student's current system prompt. The teacher then opens a multi-turn conversation with the student, whose responses are generated under that prompt. Through successive questions and follow-ups, the teacher builds a persistent diagnosis of where the student's behavior diverges from the desired behavior, attributes that divergence to a specific deficiency in the prompt, and emits a minimal, targeted edit. The conversation is discarded, the edit is applied, and a fresh diagnostic conversation begins against the repaired prompt. The loop terminates when the teacher can no longer surface a material weakness or a budget is exhausted.

DDO sits at the intersection of three research areas that have not previously been combined (see Table 1 and Section 2):

- **Conversational evaluation**, which runs multi-turn LLM–LLM dialogues to *score* agents but does not act on the result.
- **Adversarial probing / red-teaming** (e.g., PAIR [10], ATA [13]), which conducts live conversations to *expose* weaknesses but aims to break rather than repair.
- **Prompt optimization** (ProTeGi, OPRO, MARS [6]), which *repairs* prompts but evaluates in batch or in single-turn exchanges, never through genuine multi-turn diagnosis.

DDO's novelty is the closed loop *converse* $\rightarrow$ *diagnose* $\rightarrow$ *repair* $\rightarrow$ *reset* that connects live conversational diagnosis to targeted prompt repair.

Contributions.

1. We identify the *conversational-diagnosis gap* in automatic prompt optimization: existing methods either converse without repairing or repair without conversing (Section 2).
2. We introduce **DDO**, a prompt optimization framework built on a multi-turn teacher–student diagnostic loop with a persistent weakness profile and surgical prompt edits, and formalize it as a partially observable optimization process (Section 4).
3. We give a concrete instantiation—agent roles, the diagnostic policy, the weakness-profile representation, the prompt-edit operator, and termination criteria—that can be implemented on top of standard chat APIs (Section 5).
4. We specify a complete experimental protocol with baselines, ablations, and metrics over reasoning, instruction-following, and tool-use tasks (Section 6). **[TODO: Empirical results are pending and will be reported in a future revision.]**

2 Related Work

2.1 Batch-and-rewrite prompt optimization

APE [1] generates candidate instructions with an LLM and selects among them by scoring on a task. OPRO [2] treats the LLM as a black-box optimizer, feeding it a trajectory of previously tried prompts with their scores and asking for a better one. ProTeGi [3] introduces "textual gradients": it collects failing minibatch examples, asks an LLM to articulate *why* the prompt failed (the gradient), and edits in the indicated direction, using beam search and bandit-style selection to manage the query budget. PromptAgent [4] frames optimization as planning over an error-driven state space, and PromptWizard [5] uses a critique-and-synthesis loop. All of these share the defining feature that the signal driving each edit is an aggregate over a static evaluation set; the optimizer never interacts with the model turn-by-turn.

2.2 LLM-as-optimizer with dialogue framing

MARS [6] is the closest prior work in spirit: it organizes prompt optimization around a "Teacher–Critic–Student" triad using Socratic-style questioning, where a Teacher poses guiding questions, a Critic checks that the questions are genuinely Socratic, and a Student rewrites the prompt. Crucially, however, in MARS the Student *is a prompt-writing module*, not the model whose behavior is being evaluated; each exchange is a single-turn API call without persistent dialogue state; and the questions concern the abstract quality of the prompt rather than the observed behavior of a model running under it. DDO differs on all three axes: the student is the model under optimization, the diagnosis is genuinely multi-turn with carried state, and the teacher reasons from the student's *behavior* rather than from the prompt text alone.

2.3 Conversational and agentic evaluation

A growing line of work evaluates LLMs and LLM agents over multi-turn dialogues rather than single prompts: MT-Bench-style and MultiChallenge [7] benchmarks measure conversational persistence and context management; ToolSandbox [8] introduces an LLM "user simulator" that interacts with a tool-use agent across turns; and surveys document the shift toward sustained, context-aware evaluation [9]. These methods produce scores or qualitative reports; none feeds the dialogue outcome back into an automatic prompt repair.

2.4 Adversarial probing and red-teaming

PAIR [10] runs an attacker LLM in a loop against a target, iteratively refining adversarial prompts from the target's responses; TAP [11] extends this with tree search. MART [12] alternates an adversarial prompt generator with safety fine-tuning of the target across rounds. Most relevant, the Agent-Testing Agent (ATA) [13] is a meta-agent that conducts adversarial conversational tests whose difficulty adapts via judge feedback, steering subsequent conversations toward an agent's weakest capabilities and emitting bug reports. ATA is, in effect, the diagnostic front-half of DDO; but its objective is to surface and report failures, and it modifies neither the prompt nor the model. DDO can be viewed as closing ATA's loop: routing the conversational diagnosis into a targeted prompt-repair operator and then re-diagnosing.

Method	Multi-turn dialogue	Diagnoses behavior	Repairs prompt	Student = model under test
ProTeGi / OPRO / APE	✗	partial	✓	✓
MARS	✗*	✗	✓	✗
MT-Bench / ToolSandbox	✓	✓	✗	✓
PAIR / ATA	✓	✓	✗	✓
DDO (ours)	✓	✓	✓	✓

Table 1: Positioning of DDO. *MARS uses a single-turn "Socratic" exchange between optimizer modules, not a multi-turn conversation with the model under test. "✓" / "✗" denote presence/absence of the property.

2.5 Knowledge distillation and student-aware teaching

Teacher–student framings are pervasive in distillation, where a teacher's outputs supervise a student model's weights [14]. Student-aware variants such as PromptKD [15] tune soft prompts so the teacher produces student-friendly targets, and preference-aligned teaching (e.g., ARTE) generates data matched to student weaknesses. DDO borrows the *student-aware* intuition—adapt to the specific learner's deficiencies—but operates at the level of a discrete, human-readable system prompt and via natural-language diagnosis rather than gradient distillation, and it modifies the prompt rather than the weights.

2.6 Summary of the gap

Table 1 positions DDO. To our knowledge, no prior method combines (a) a genuine multi-turn diagnostic conversation with the model under optimization, (b) a persistent cross-turn weakness profile, and (c) a targeted prompt-repair operator closing the loop.

3 Problem Formulation

Let $\mathcal{S}_\theta$ denote a student LLM with fixed weights, operating under a natural-language system prompt p drawn from the space $\mathcal{P}$ of admissible prompts. For a task $\mathcal{T}$ with input distribution $\mathcal{D}$ and a scalar quality metric $Q(\cdot)$, the goal of prompt optimization is

$$p^\star = \arg\max_{p\in\mathcal{P}} \mathbb{E}_{x\sim\mathcal{D}}\big[\, Q\big(\mathcal{S}_\theta(x;p),\, x\big)\,\big]. \tag{1}$$

Batch-and-rewrite methods approximate the expectation in (1) with a fixed evaluation set $\mathcal{D}_{\text{eval}}$ and use the aggregate $\hat{Q}(p) = \frac{1}{|\mathcal{D}_{\text{eval}}|}\sum_x Q(\mathcal{S}_\theta(x;p), x)$ as the sole feedback signal at each step.

DDO instead introduces a teacher $\mathcal{S}_\phi$ (typically a stronger model, $\phi \neq \theta$) and a behavioral specification β—a natural-language description of the task-relevant axes of behavior the teacher should probe (e.g., multi-step reasoning fidelity, instruction adherence, output format, refusal calibration, edge-case handling). The teacher's role is to recover, from interaction alone, the direction in prompt space along which p most underperforms with respect to β.

Definition 1 (Diagnostic conversation)**.** A diagnostic conversation of horizon H is a sequence $c = (u_1, a_1, u_2, a_2, \ldots, u_H, a_H)$ where each teacher utterance $u_t = \pi_\phi(\beta, p, c_{<t})$ is generated by the teacher's diagnostic policy π_ϕ conditioned on the specification β, the student's current prompt p, and the conversation so far $c_{<t}$; and each student response $a_t = \mathcal{S}_\theta(u_t; p, c_{<t})$ is generated by the student under prompt p with the running conversation as context.

Definition 2 (Weakness profile)**.** A weakness profile $w \in \mathcal{W}$ is a structured summary, produced by the teacher from a completed conversation, $w = \rho_\phi(\beta, p, c)$, recording (i) which axes of β the student handled adequately, (ii) the single most salient deficiency observed, (iii) the evidential turns supporting that diagnosis, and (iv) a confidence estimate.

Definition 3 (Prompt-repair operator)**.** A prompt-repair operator $\mu_\phi : \mathcal{P} \times \mathcal{W} \to \mathcal{P}$ maps the current prompt and a weakness profile to a revised prompt, $p' = \mu_\phi(p, w)$, under a minimality constraint: p' should differ from p only in the manner required to address the salient deficiency recorded in w.

Problem 1 (Diagnostic Dialogue Optimization)**.** Given $\mathcal{S}_\theta$, teacher $\mathcal{S}_\phi$, specification β, initial prompt p_0, and a budget B (e.g., total teacher–student exchanges), find a sequence $p_0 \to p_1 \to \cdots \to p_K$ with $p_{k+1} = \mu_\phi(p_k, \rho_\phi(\beta, p_k, c_k))$, where c_k is a diagnostic conversation against p_k, such that $\hat{Q}(p_K)$ is maximized subject to $\sum_k |c_k| \leq B$.

A partially observable view. DDO is naturally cast as a partially observable decision process. The hidden state is the true behavioral profile of $\mathcal{S}_\theta$ under p, which is never directly observable. Each teacher utterance is an *information-gathering action* that yields an observation (the student's response), narrowing the teacher's posterior over where the prompt is deficient. The repair operator is a *control action* that changes the prompt—and hence the hidden behavioral state—after which the posterior is reset and re-estimated. This framing clarifies why a persistent within-conversation profile matters: the value of the t-th question depends on what the first $t-1$ answers have already revealed, exactly as in optimal experimental design.

4 Method: Diagnostic Dialogue Optimization

DDO instantiates the formulation above with four cooperating components and an outer control loop.

4.1 Components

Teacher ($\mathcal{S}_\phi$). A capable model carrying the behavioral specification β in its own system prompt. It plays two roles: *interrogator* (generating u_t via π_ϕ) and *diagnostician* (producing w via ρ_ϕ and the edit via μ_ϕ). These can be the same underlying model with different instructions, or separated for modularity.

Student ($\mathcal{S}_\theta$). The model under optimization, run with the current candidate prompt p_k as its system prompt and the live conversation as context. The student is never told it is being evaluated, to avoid eliciting unrepresentative "test-taking" behavior.

Verifier (optional). A lightweight scorer $\hat{Q}$ over a small validation set, queried only between outer iterations to decide whether to accept a repair (a guardrail against regressions; see Section 5).

Memory. A persistent store holding the current prompt, the running conversation $c_{<t}$ within an iteration, and the history of accepted weakness profiles and edits across iterations (used to avoid re-diagnosing already-fixed issues and to detect oscillation).

Algorithm 1 Diagnostic Dialogue Optimization (DDO)

Require: student $\mathcal{S}_\theta$, teacher $\mathcal{S}_\phi$, spec β, initial prompt p_0, horizon H, budget B, patience m

$p \leftarrow p_0$; best $\leftarrow p_0$; hist $\leftarrow \emptyset$; spent $\leftarrow 0$; stall $\leftarrow 0$
while spent $< B$ **and** stall $< m$ **do**
  $c \leftarrow ()$ ▷ fresh conversation against current prompt
  for $t = 1$ **to** H **do**
    $u_t \leftarrow \pi_\phi(\beta, p, c, \text{hist})$ ▷ adaptive diagnostic question
    $a_t \leftarrow \mathcal{S}_\theta(u_t; p, c)$ ▷ student answers under prompt p
    $c \leftarrow c \,\|\, (u_t, a_t)$; spent $\leftarrow$ spent $+1$
    if π_ϕ signals diagnosis complete **then break**
    end if
  end for
  $w \leftarrow \rho_\phi(\beta, p, c)$ ▷ compile weakness profile
  if w.confidence $< \tau$ **or** w reports no material weakness **then**
    stall $\leftarrow$ stall $+1$; **continue**
  end if
  $p' \leftarrow \mu_\phi(p, w)$ ▷ minimal targeted edit
  if verifier present **and** $\hat{Q}(p') < \hat{Q}(p) - \epsilon$ **then**
    stall $\leftarrow$ stall $+1$ ▷ reject regression, keep p
  else
    $p \leftarrow p'$; hist $\leftarrow$ hist $\cup \{w, \text{diff}(p, p')\}$; stall $\leftarrow 0$
    if verifier present **and** $\hat{Q}(p) > \hat{Q}(\text{best})$ **then** best $\leftarrow p$
    end if
  end if
end while
return verifier present **?** best **:** p

4.2 The DDO loop

Algorithm 1 gives the procedure. Within an outer iteration, the teacher conducts a bounded diagnostic conversation, choosing each question adaptively from the answers so far and steering toward whichever axis of β looks weakest. It then compiles a weakness profile and proposes a minimal edit. An optional verifier gates the edit on a small validation set to prevent regressions; accepted edits update the prompt, and the conversation is discarded so the next iteration probes the *repaired* prompt from scratch. The loop halts when no material weakness is found, improvements stall, or the budget is exhausted.

4.3 Why diagnosis beats aggregate error

Batch methods compress all behavioral information into a scalar (or a few sampled failures) before the optimizer ever reasons about the prompt. DDO preserves the *causal trace*: because the teacher chose each question and saw each answer, the weakness profile is grounded in a specific, reproducible interaction rather than a post-hoc guess about why an aggregate score is low. Two consequences follow. First, the edit can be *minimal and targeted*—it names the exact behavior to change—reducing the prompt bloat and over-fitting that accompany repeated whole-prompt rewrites. Second, the method is *label-light*: a diagnostic conversation can reveal a reasoning or formatting deficiency without a labeled dataset for that deficiency, because the teacher supplies the judgment online. We make these claims precise as testable hypotheses in Section 6.

5 Instantiation

We describe concrete design choices used in our reference implementation. All components are realizable with standard chat-completion APIs; no model weights are modified.

5.1 Behavioral specification β

β is a short rubric enumerating the axes the teacher should probe, with one line each, e.g.: *stepwise reasoning* (does the student decompose before answering?), *instruction adherence* (does it honor explicit constraints?), *format* (does output match the required schema?), *robustness* (does it handle under-specified or adversarial phrasing?), and *calibration* (does it hedge or refuse appropriately?). β encodes the teacher's prior over "what matters" and is the only task-specific human input besides the initial prompt.

5.2 Diagnostic policy π_ϕ

The teacher is instructed to (i) begin broad, sampling one question per axis of β; (ii) when a response shows weakness, *drill down* with follow-ups on that axis before moving on; (iii) avoid leading questions that reveal the answer; and (iv) emit a special token when it has localized a single dominant deficiency or exhausted the horizon H. Carrying $c_{<t}$ in context is what makes the policy adaptive: the t-th question is conditioned on all prior answers.

5.3 Weakness profile ρ_ϕ and edit μ_ϕ

ρ_ϕ outputs a structured object (we use JSON) with fields `axes_ok`, `salient_deficiency`, `evidence_turns`, `hypothesized_prompt_cause`, and `confidence`. μ_ϕ is instructed to produce the smallest prompt change that plausibly fixes `salient_deficiency`, preferring to *add a targeted instruction or a single clarifying example* over rewriting existing text, and to return a unified diff alongside the new prompt so edits are auditable and reversible.

5.4 Guardrails

Three guardrails stabilize the loop. (1) A *verifier* on a small validation set rejects edits that regress aggregate quality, converting DDO into a hill-climber with diagnostic proposals. (2) A *history check* forbids re-fixing an axis already marked resolved unless new evidence contradicts it, preventing oscillation. (3) A *minimality penalty* discourages the prompt from growing without bound across iterations.

5.5 Cost accounting

The dominant cost is teacher–student exchanges. With horizon H and K outer iterations, DDO uses $O(KH)$ student calls for diagnosis plus optional verifier calls; this is directly comparable to the minibatch evaluations consumed by ProTeGi/OPRO and is the quantity we hold fixed in budget-matched comparisons (Section 6).

6 Experimental Protocol

This section specifies the planned evaluation. Numerical results are placeholders to be filled after the experiments are run.

6.1 Hypotheses

H1 (Quality). At a matched query budget, DDO reaches higher task quality than batch-and-rewrite baselines.

H2 (Efficiency). DDO attains a given quality level with fewer student queries, because diagnostic questions are more informative per call than random minibatch evaluations.

H3 (Label efficiency). DDO's advantage grows as labeled data shrinks, because the teacher supplies online judgments.

H4 (Edit locality). DDO produces shorter, more interpretable prompt diffs than whole-prompt rewriters at equal quality.

6.2 Tasks and datasets

We propose three families: (i) *reasoning*—BIG-Bench Hard subtasks and GSM8K-style arithmetic word problems; (ii) *instruction following*—a multi-constraint following suite (e.g., IFEval-style constraints); (iii) *tool use / agentic*—a ToolSandbox-style environment requiring correct multi-turn tool calls. **[TODO: Finalize the exact dataset versions, splits, and licenses.]**

6.3 Models

Student models spanning a capability range (e.g., a small open-weight model and a mid-size model) optimized under each method; a single stronger teacher held fixed across methods to ensure fair comparison. **[TODO: Specify exact model identifiers and versions.]**

6.4 Baselines

(1) Zero-shot single-rewrite (ask the teacher once to improve the prompt); (2) APE; (3) OPRO; (4) ProTeGi; (5) MARS. All run at the same total query budget B as DDO.

6.5 Ablations

(A) *No memory*: reset the within-conversation profile each turn (tests the value of persistent diagnosis). (B) *Single-turn*: set $H = 1$ (collapses DDO toward MARS-style exchange). (C) *No verifier*: remove the regression gate. (D) *No minimality*: allow whole-prompt rewrites in μ_ϕ. (E) *Spec sensitivity*: vary the axes in β.

6.6 Metrics

Task quality $\hat{Q}$ on held-out test sets; queries-to-target (for H2); quality vs. label-budget curves (for H3); prompt edit distance and final prompt length (for H4); and qualitative analysis of representative diagnostic transcripts and the edits they induced.

Method	Reasoning	Instr.-follow	Tool use
Zero-shot rewrite	-.-	-.-	-.-
APE	-.-	-.-	-.-
OPRO	-.-	-.-	-.-
ProTeGi	-.-	-.-	-.-
MARS	-.-	-.-	-.-
DDO (ours)	-.-	-.-	-.-

Table 2: **[TODO: Main results at matched query budget. Fill with held-out accuracy / task metric once experiments are complete.]**

6.7 Results

[TODO: Add ablation table, queries-to-target plot, label-efficiency curves, edit-locality statistics, and qualitative transcript analysis.]

7 Discussion

Relation to optimal experimental design. DDO's inner loop is a form of active information gathering: each question is chosen to maximally reduce uncertainty about where the prompt is deficient. This suggests principled extensions—e.g., explicit value-of-information criteria for question selection, or Thompson-sampling over axes of β.

When DDO should and should not help. We expect the largest gains where weaknesses are *behavioral and localizable*—a missing instruction, an unhandled edge case, an ambiguous format—and conversation can expose them quickly. We expect smaller gains where the bottleneck is the student's raw capability (no prompt fixes it) or where the task is so narrow that a handful of labeled examples already pinpoint the failure, neutralizing the conversational advantage.

Teacher dependence. DDO's diagnoses are only as good as the teacher's judgment. A weak or miscalibrated teacher may hallucinate weaknesses or propose harmful edits; the verifier guardrail bounds the downside but does not eliminate it. Studying robustness to teacher quality is part of the planned ablations.

8 Limitations

DDO assumes API access to both models and incurs teacher cost per diagnostic turn. The method optimizes a discrete system prompt and does not modify weights, so it cannot overcome fundamental capability gaps. Diagnoses are natural-language judgments and inherit the teacher's biases. Finally, the empirical claims in this draft are *hypotheses*; the results tables are placeholders pending experiments. **[TODO: Update this section once results are available.]**

9 Conclusion

We introduced Diagnostic Dialogue Optimization, which replaces the batch-and-rewrite loop of automatic prompt optimization with a multi-turn teacher–student diagnostic conversation that localizes a prompt's weakness and repairs it with a minimal, targeted edit before re-diagnosing from scratch.

DDO unifies the diagnostic strength of conversational red-teaming with the repair capability of prompt optimizers, occupying a gap left open by both. We provided a formalization, an algorithm, a concrete instantiation, and a full experimental protocol; completing the empirical evaluation is the immediate next step.

Reproducibility Statement

[TODO: On acceptance/release, provide a code repository, the exact prompts for β, π_ϕ, ρ_ϕ, and μ_ϕ, model versions, dataset splits, and random seeds. Release representative diagnostic transcripts and the induced prompt diffs.]

Ethics Statement

DDO is a capability-improvement method for prompts and shares the dual-use considerations of prompt optimization and automated red-teaming generally: the same diagnostic loop that repairs a benign prompt could in principle be aimed at probing safety guardrails. We intend the method for improving task reliability and recommend that practitioners pair it with safety evaluation rather than using the diagnostic component in isolation to weaken safeguards.

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